

Homework Stochastic Simulation - Third set

For the programming exercises, you are free to use any programming language or mathematical software package (Maple, Matlab, Mathematica, R, anything) as long as you

- include all code in your deliverables, and
- for random number generation purposes, you use standard built-in methods to generate standard uniform samples. For generating samples from any other distribution, you'll have to construct them yourself using the standard uniform samples.

You are expected to submit the homework individually. Please compile answers and code in a single PDF document and upload it to canvas. For questions, you can contact Tygo Nijsten at t.g.c.nijsten@uva.nl.

Exercise 1 Make exercise V.1.4 of [AG].

Exercise 2 Make exercise V.1.5 of [AG].

Exercise 3 The number of casualty insurance claims that will be made to a branch office next week depends on an environmental factor U . If the value of this factor is $U = u$, then the number of claims will have a Poisson distribution with mean $\frac{15}{0.5+u}$. Assuming that U is standard-uniformly distributed, let p denote the probability that there will be at least 20 claims next week.

1. Explain how to obtain the crude Monte Carlo estimator of p .
2. Develop an efficient simulator that uses conditional expectation along with a control variate.
3. Develop an efficient simulator that uses conditional expectation along with antithetic variables.
4. Write a program to determine the variance of these three estimators. Which one of the methods is best?

Exercise 4 Suppose we want to use simulation to determine $\mathbb{E}[h(Z_1, \dots, Z_n)]$, where $Z_1, \dots, Z_n$ are independent standard normal variables, and where h is an increasing function of each of its coordinates. Let $W = \sum_{i=1}^n a_i Z_i$, where all the a_i are non-negative. Explain how we can use stratified sampling, stratifying on W , to approximate $\mathbb{E}[h(Z_1, \dots, Z_n)]$. Assume that the inversion method is used to sample W .

You may use that if a standard normal random variable Z is independent of a random variable X , which is normally distributed with mean μ and variance σ^2 , then the conditional distribution of Z given that $X + Z = t$ is normal with mean $\frac{t-\mu}{1+\sigma^2}$ and variance $\frac{\sigma^2}{1+\sigma^2}$.

Exercise 5 Make exercise V.7.1 of [AG], by interpreting the expression as $\mathbb{E}[h(X)]$ for some function h , and some random variable X . Suggest at least three methods, of which one is based on control variates and one is based on stratification (the third is completely up to you!). Explain how of each of the methods work, and explain why variance reduction is to be expected in each of them.